EXPERIMENT :17

Simulation of V2V and V2I Latency and Packet Delivery

OBJECTIVES :

• To model a simple Vehicle-to-Vehicle (V2V) or Vehicle-to-Infrastructure (V2I) communication scenario using MATLAB.

• To estimate communication latency and packet delivery ratio (PDR) under different vehicle distances and traffic/channel conditions.

• To compare the communication performance of at least two V2X scenario settings.

• To analyze the relationship between distance, latency, packet delivery, and V2X safety requirements.

PROGRAME CODE :

```
clc;
clear;
close all;

%% 1. SIMULATION PARAMETERS

N = 1000;                   % Total number of packets
packetSize = 300*8;         % Packet size in bits
Rb = 10e6;                  % Data rate (10 Mbps)

c = 3e8;                    % Speed of light (m/s)

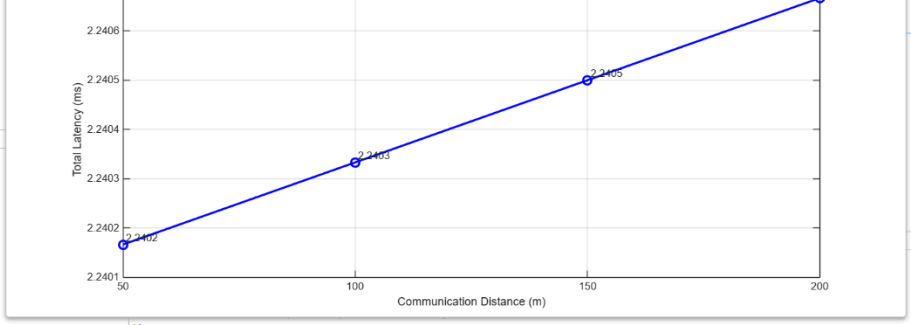
Tp = 1e-3;                  % Processing delay (seconds)
Tq = 1e-3;                  % Queuing delay (seconds)

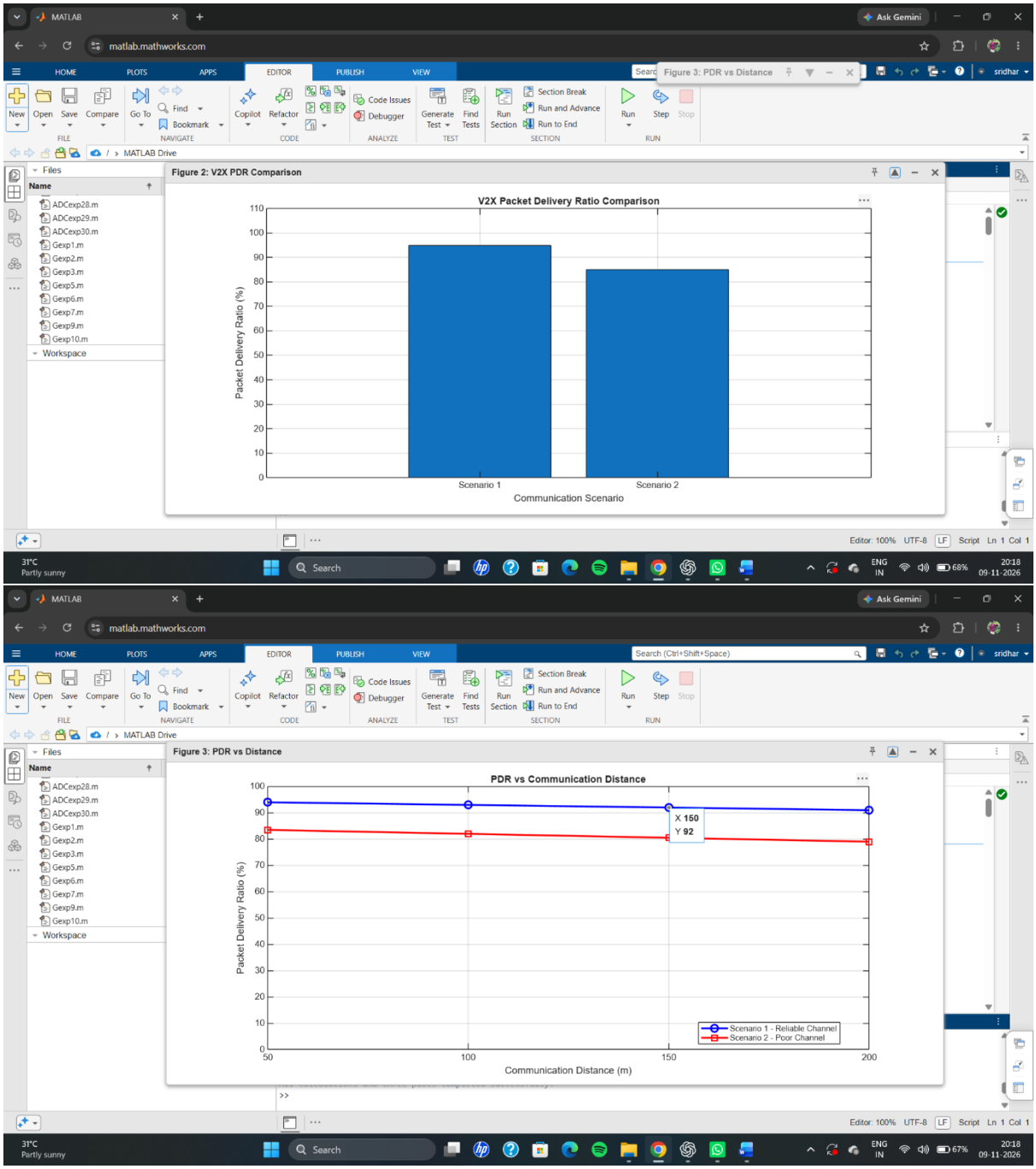
% Communication distances
distance = [50 100 150 200];

% Packet loss probabilities
```

```
Scenario 2 PDR  : 85.00 %
Scenario 1 Lost Packets : 50
Scenario 2 Lost Packets : 150

All calculations and three plots completed successfully.
>>
```





RESULT : The MATLAB simulation of V2X communication was successfully implemented and analyzed based on communication latency, Packet Delivery Ratio (PDR), and communication distance. The results showed that Scenario 1 achieved a higher PDR of 95%, while Scenario 2 achieved 85%. The total communication latency increased slightly with distance, from approximately 2.2402 ms to 2.2407 ms. The simulation demonstrates that reliable communication channels provide better packet delivery performance and that communication distance affects propagation delay and the modeled PDR.